

XII Latin-American Algorithms, Graphs and Optimization Symposium (LAGOS 2023)

Recovering cyclic tilings through β -skeletonsHéctor Saib Maravillo Gómez^a, Gilberto Calvillo Vives^a, Erick Treviño Aguilar^{a,*}^a*Institute of Mathematics, National Autonomous University of Mexico, Cuernavaca, Mexico*

Abstract

There is a class of proximity graphs parameterized by a real number β that are called β -skeletons and which are geometrical graphs whose vertex set V is a discrete set of points in the plane and two points p and q are adjacent if certain region $\text{lune}_\beta(p, q)$ defined by them does not contain any other vertex. Special cases of β -skeletons are the Gabriel Graph ($\beta = 1$) and the Relative Neighborhood Graph ($\beta = 2$). Cyclic tilings are tilings of the plane whose tiles are convex cyclic polygons that contain the center of the circle in which they are inscribed. Up to our knowledge, this class of tilings has not been studied before. They generalize regular, Archimedean, and k -uniform tilings. We give sufficient conditions for the graph of a cyclic tiling to be a β -skeleton of its vertices. The result is particularly nice for the case of the Gabriel graph. The study of the relationship between tilings arises while studying street networks. This relationship is illustrated briefly.

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Peer-review under responsibility of the scientific committee of the XII Latin-American Algorithms, Graphs and Optimization Symposium

Keywords: Proximity graph; β -skeleton; relative neighborhood graph; Gabriel graph; cyclic tiling; street network

1. Introduction

Understanding urban street networks is relevant because it allows for comparing the mobility capacities of cities, identifying network vulnerabilities, and improving urban and transport planning; see [1]. Street networks are represented by graphs whose vertices are the intersections and endpoints of the streets, and the edges correspond to street segments; see [2]. If tunnels and overpasses are ignored, by construction, it is a planar graph.

Many studies characterize street networks through measurements and distributions of topological and geometric variables of street networks (see [3, 4] for a general survey). An alternative approach, less explored, is to approximate street networks using proximity graphs. In these graphs, given a set of points in the plane, two points are joined by an edge if a region defined by them does not contain any other point in the set. Some examples are the Gabriel graph (GG), the β -skeleton, and the relative neighborhood graph (RNG), defined in Section 2. Empirical studies showed that the β -skeleton closely approximates the street networks of some cities in the United States and Japan; see [5, 6]. Besides, the β -skeletons reproduce statistical properties of the street networks of 269 cities in the world; see [7]. For

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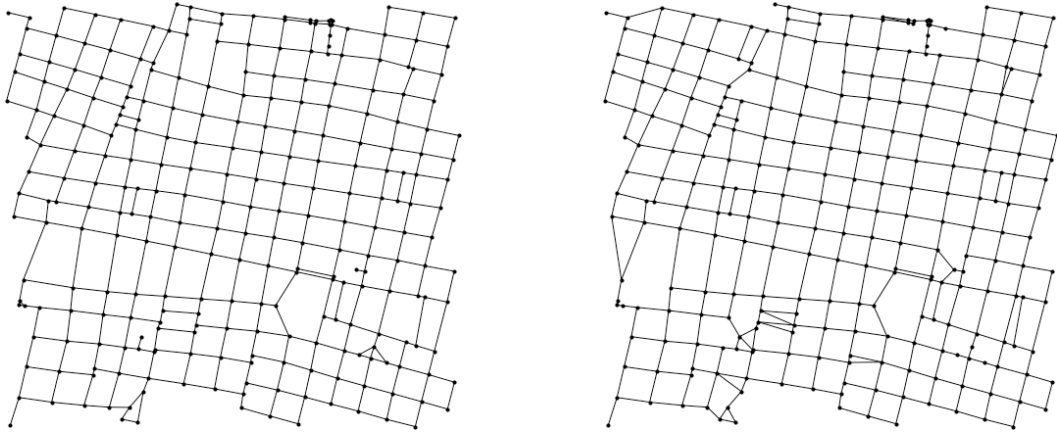


Fig. 1. Street network of the center of Mérida city, Mexico, and its β -skeleton, with $\beta = 1.4$.

example, Figure 1 shows the street network of the center of Mérida city, Mexico, and a β -skeleton generated by its vertices. Distinguishing which is the street network and which is the β -skeleton is not easy.

To understand how some proximity graphs approximate street networks, we address the ideal case where street networks are tilings of the plane made up of convex polygons. We found that a tiling is the β -skeleton of its vertices when the central angle of the edges of their polygons is not too large.

2. Definitions

With $B(c, r)$ and $C(c, r)$, we denote the open disk and the circle with center $c \in \mathbb{R}^2$ and radius $r > 0$, respectively. For $\beta \geq 1$ a real number, we define:

$$c_1 = \frac{\beta}{2}p + \left(1 - \frac{\beta}{2}\right)q, \quad (1)$$

$$c_2 = \left(1 - \frac{\beta}{2}\right)p + \frac{\beta}{2}q, \quad (2)$$

$$r_\beta = \frac{\beta}{2}d(p, q), \quad (3)$$

$$\text{lune}_\beta(p, q) = B(c_1, r_\beta) \cap B(c_2, r_\beta). \quad (4)$$

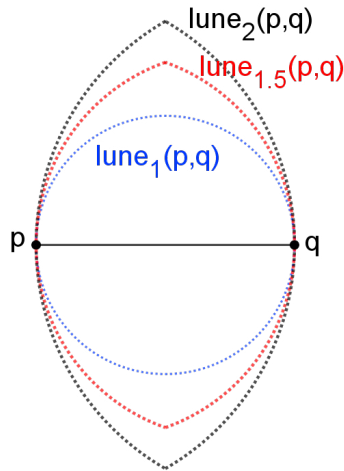
See examples of $\text{lune}_\beta(p, q)$ for three values of β in Figure 2.

Definition 1 (β -skeleton [8]). Let V be a discrete set¹ of points in $\mathbb{R}^2$. The β -skeletons of V are a family of graphs parameterized by $\beta \geq 1$, denoted $G_\beta(V) = (V, E_\beta)$. Let p and q be disjoint vertices in V and $d(p, q)$ the Euclidean distance between them, then:

$$(p, q) \in E_\beta \Leftrightarrow \text{lune}_\beta(p, q) \cap V = \emptyset. \quad (5)$$

The 1-skeleton is called the Gabriel graph (GG) and the relative neighborhood graph (RNG) is the 2-skeleton. By its definition, $RNG(V) \subset G_{\beta_1} \subset G_{\beta_2} \subset GG(V)$ for $2 \geq \beta_1 \geq \beta_2 \geq 1$. If V is finite, it was shown that the Euclidean

¹ Originally, β -skeletons were defined for finite sets of points, but the definition can be extended to discrete sets, possibly of infinite cardinality, with no changes.

Fig. 2. $\text{lune}_\beta(p, q)$ for $\beta = 1, 1.5$ and 2 .

minimum spanning tree of V (a connected graph) is a subgraph of $\text{RNG}(V)$ [9, see Theorem 3.1]. Also, if the points are in general position, then $\text{GG}(V)$ is a subgraph of the Delaunay triangulation of V , which is a planar graph [10, see Lemma 2]. These properties ensure that β -skeletons are planar and connected graphs for $1 \leq \beta \leq 2$.² See [12, 13, 14] for a general survey about proximity graphs and their properties.

A *tiling* or *tessellation* of the plane is a countable family of closed sets with non-empty interior $T = \{T_1, T_2, \dots\}$, called tiles, that cover the plane, and their interiors are pairwise disjoint. If the tiles of T are polygons, the vertices and edges of T are the vertices and edges of its tiles. We denote edges of T with E_T and its vertices with V . A tiling of this sort is *edge-to-edge* if the intersection between two tiles T_i and T_j is one of the following options: 1) the empty set, 2) a vertex of T_i and T_j , or 3) an edge of T_i and T_j ; see [15].

Definition 2. Let $T = \{T_1, T_2, \dots\}$ be an edge-to-edge tiling, with a discrete set of vertices. We define T to be *cyclic* if each tile T_i is a convex polygon inscribed in a circle $C_i = C(c_i, r_i)$ and the center c_i is in the interior of T_i .

Regular, Archimedean, and k -uniform tilings are special cases of cyclic ones; see definitions and examples in [15]. We show three examples in Figure 3.

3. Results

We say that a proximity graph $G = (V, E)$ *recovers* a tiling T with its set of vertices V and its set of edges E_T , if the following equivalence holds true:

$$(p, q) \in E \Leftrightarrow \overline{pq} \in E_T. \quad (6)$$

The relative neighborhood graph recovers the three regular tilings: triangular, square, and hexagonal; see [9]. We generalize this fact by proving a theorem that establishes sufficient conditions under which a β -skeleton recovers cyclic tilings. The condition relates the maximum central angle of the edges of the tiles T_i for C_i and β . The proof of the theorem uses three lemmas that are stated in the final part of this section.

Theorem 1. Let $T = \{T_1, T_2, \dots\}$ be a cyclic tiling, V its set of vertices and E_T its set of edges. For $\beta \geq 1$, let $G_\beta(V) = (V, E_\beta)$ be the β -skeleton of V , then $E_\beta \subseteq E_T$.

² In fact, $\text{lune}_2(p, q)$ is the maximal region that ensures the connectedness of the graph, in the sense that any proximity graph defined by a region (connected and symmetric for p and q) is included in $\text{lune}_2(p, q)$ is connected. Furthermore, the closure of $\text{lune}_1(p, q)$ is a minimal region for the planarity of the graph; see [11].

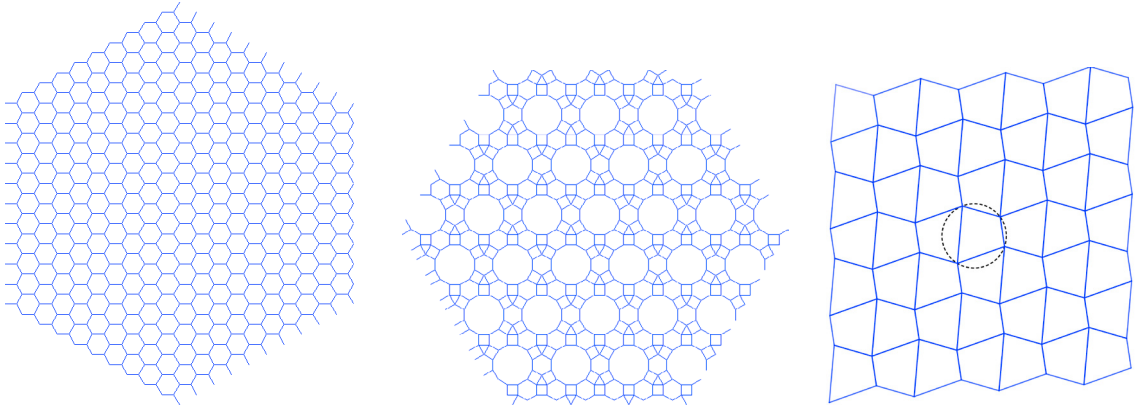


Fig. 3. Examples of cyclic tilings: regular (6^3), k -uniform (3.4.6.12) (source [16]), and cyclic non-uniform (own elaboration).

Let $C_i = C(c_i, r_i)$ be the circle that circumscribes the corresponding tile T_i , and let α be the supremum of central angles of the edges of T . Then, the Gabriel graph of V recovers T . Furthermore, if

$$\left(\frac{1 + \cos(\frac{\alpha}{2})}{\sin(\frac{\alpha}{2})} \right)^2 + 1 \geq 2\beta, \quad (7)$$

holds true for some $\beta > 1$, then $G_\beta(V)$ recovers T .

Proof. The idea of the proof is to show the double inclusion, between the edges E_β and the edges E_T . For the inclusion, $E_T \subseteq E_\beta$, we first show that for an edge $\overline{pq}$ that belongs to two tiles T_i and T_j , the $\text{lune}_\beta(p, q)$ is included in the union of the interiors of C_i and C_j if inequality (7) holds. We remark that this inequality always holds for $\beta = 1$, that is, for the Gabriel graph. Furthermore, we prove that no vertex of T is in the interior of any C_i . Therefore, all the edges of T are in $G_\beta(V)$. Then, we show that any pair of vertices that do not form an edge of T is not an edge of $G_\beta(V) = (V, E_\beta)$, thus, $E_\beta \subseteq E_T$.

We start with the inclusion $E_T \subseteq E_\beta$. Let $\overline{pq}$ be an edge of T that belongs to tiles T_i and T_j , as in Figure 4. We denote with α_i and α_j the central angles of $\overline{pq}$ in C_i and C_j , respectively.

Note that the function

$$f(x) := \left(\frac{1 + \cos(x)}{\sin(x)} \right)^2 + 1, \quad (8)$$

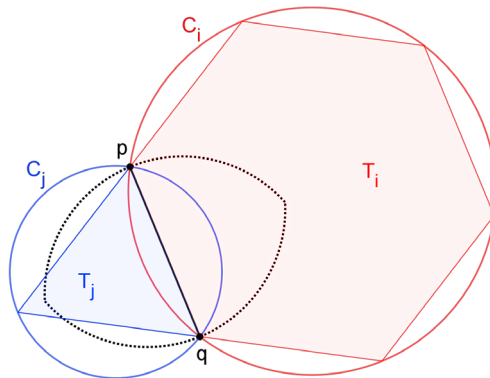


Fig. 4. Two adjacent tiles through $\overline{pq}$ and $\text{lune}_{1.5}(p, q)$.

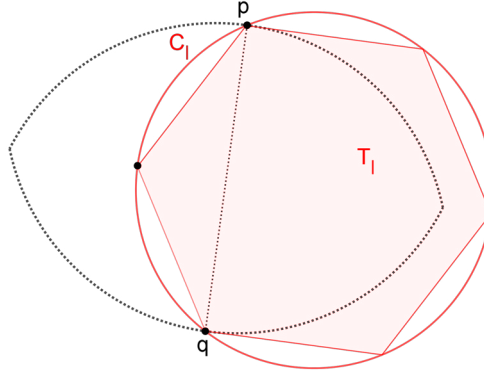


Fig. 5. Two non-successive vertices p and q of a tile T_l and $\text{lune}_{1,5}(p, q)$.

is continuous, positive, and strictly decreasing in $x \in (0, \pi]$. Then

$$\min \left\{ f\left(\frac{\alpha_i}{2}\right), f\left(\frac{\alpha_j}{2}\right) \right\} \geq f\left(\frac{\alpha}{2}\right) \geq f\left(\frac{\pi}{2}\right) = 2. \quad (9)$$

Therefore, the inequality (7) always holds true for $\beta = 1$. Given α , let β be such that $f\left(\frac{\alpha}{2}\right) \geq 2\beta$. Then, by Lemma 1, we have that:

$$\text{lune}_\beta(p, q) \subset B(c_i, r_i) \cup B(c_j, r_j). \quad (10)$$

Since each tile T_i is a cyclic polygon and contains the center c_i of the circumscribing circle, then by Lemma 3 there is no vertex of T inside any $B(c_i, r_i)$. Therefore,

$$\text{lune}_\beta(p, q) \cap V = \emptyset, \quad (11)$$

hence, (p, q) is an edge of $G_\beta(V)$, showing the desired inclusion $E_T \subseteq E_\beta$.

Now, we show that $\overline{pq} \notin E_T \implies (p, q) \notin E_\beta$. We start with the case in which p and q are non-successive vertices of the same tile T_l , namely, the segment $\overline{pq}$ is a diagonal of T_l ; see Figure 5. Note that in this case, $\overline{pq}$ is not an edge of E_T . Then there is at least one more vertex on each of the arcs that define the diagonal $\overline{pq}$, see again Figure 5. This implies that one of these vertices lies within $\text{lune}_\beta(p, q)$ for any $\beta \geq 1$. That is, $(p, q) \notin E_\beta$.

Let p and q be vertices of T that do not belong to the same tile, as in Figure 6, and let w be the intersection between the segment $\overline{pq}$ and the boundary of $K := \cup_{i=1}^n T'_i$, where $T'_1, \dots, T'_n$ are the tiles to which p belongs to.³ If w belongs to V and since $w \in \text{lune}_\beta(p, q)$, then $\text{lune}_\beta(p, q) \cap V \neq \emptyset$, thus $(p, q) \notin E_\beta$.

Suppose now that $w \notin V$. Then, w is in an edge $\overline{st}$ of the tile $T_l \subset K$ and $w \neq s, t$. Let z be the intersection between $\overline{pq}$ and the boundary of C_l . Since p is a vertex of T_l , then the chord $\overline{pz}$ divides the circumference of C_l in two arcs, such that s is in one, and t is in the other. This implies that s or t are elements in $\text{lune}_\beta(p, z)$ for any $\beta \geq 1$. In consequence if z is in $\overline{pq}$, then for any $\beta \geq 1$:

$$\text{lune}_\beta(p, z) \subset \text{lune}_\beta(p, q), \quad (12)$$

thus, $(p, q) \notin E_\beta$.

Therefore, if two vertices of T do not form an edge in T , neither do they in G_β regardless of the value of β . Thus, $E_\beta \subseteq E_T$, for any $\beta \geq 1$. $\square$

Note that if β is such that inequality (7) holds true, the edges of T (a cyclic tiling) are edges in E_β . The inequality (7) is sufficient in order that the β -skeleton recovers T , but it is not necessary. For example, the β -skeletons always

³ The boundary of K is a simple closed curve. By the Jordan Curve Theorem, the boundary divides the plane into two disjoint connected regions. By construction, p is on the interior of K , and since q does not belong to the same tile as p , it is on the exterior of K . Thus, $\overline{pq}$ crosses the boundary of K at a vertex or edge of a tile to which p belongs.

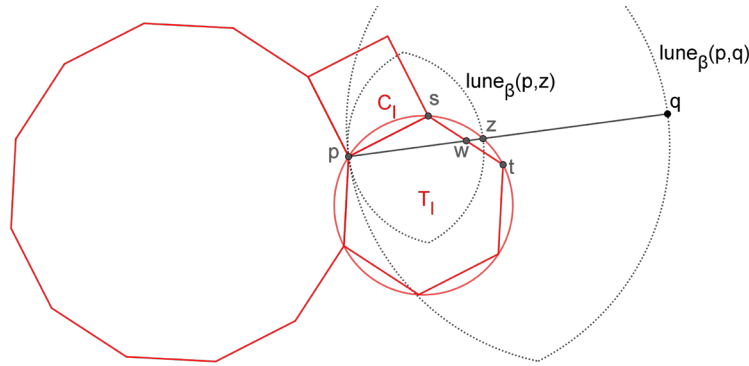


Fig. 6. Two vertices p and q that do not belong to the same tile.

recover rectangular tilings (which are cyclic tilings), although for each $\beta > 1$ we can make the α get as close as we want to π and violate equation (7).

Corollary 1.1. *Let T be a k -uniform tiling and V its set of vertices. Then, the relative neighborhood graph $RNG(V)$ recovers T .*

Proof. Since $f(x)$, defined in (8) is strictly decreasing in $(0, \pi]$, then if $x \leq \frac{\pi}{3}$:

$$f(x) \geq f\left(\frac{\pi}{3}\right) = \left(\frac{1 + \frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)^2 + 1 = 4. \quad (13)$$

Therefore, if $\beta = 2$, the inequality (7) always holds if $\frac{\alpha}{2} \leq \frac{\pi}{3}$.

In a k -uniform tiling all its tiles are regular polygons, therefore the maximum central angle α of its edges is always less than or equal to 120° . Applying Theorem 1, we conclude that $RNG(V)$ recovers T . $\square$

The following geometric lemmas are useful to better understand the proof of Theorem 1. We introduce the following notation: if $\overline{pq}$ is a chord (not diameter) of $C = C(c, r)$, the shortest arc defined by $\overline{pq}$ is denoted by $\widehat{pq}^-$, and the longest arc by $\widehat{pq}^+$. By convention p and q are neither in $\widehat{pq}^-$ nor in $\widehat{pq}^+$. Given a straight line L and a point $z \notin L$, we denote by $H(L, z)$ the open half-plane defined by L which contains z . We use the symbol $\triangle$ to denote a triangle defined by its three vertices.

Part A of Lemma 1 is used in Theorem 1 to prove Equation (10). Part B of this Lemma says that if two vertices p and q are a diagonal of a cyclic polygon, then there is a vertex of the polygon (different from p and q) inside $\text{lune}_\beta(p, q)$, for any $\beta \geq 1$, implying this diagonal is not in E_β .

Lemma 1. *Let $\overline{pq}$ be a chord (not diameter) of a circle $C = C(c, r)$, α its central angle and $\beta \geq 1$.*

A) *If*

$$\left(\frac{1 + \cos(\frac{\alpha}{2})}{\sin(\frac{\alpha}{2})}\right)^2 + 1 \geq 2\beta, \quad (14)$$

holds true, then $\text{lune}_\beta(p, q) \cap H(M, c) \subset B(c, r)$, where M is the line through the points p and q .

B) *For $\beta \geq 1$*

$$\widehat{pq}^- \subset \text{lune}_\beta(p, q). \quad (15)$$

Proof.

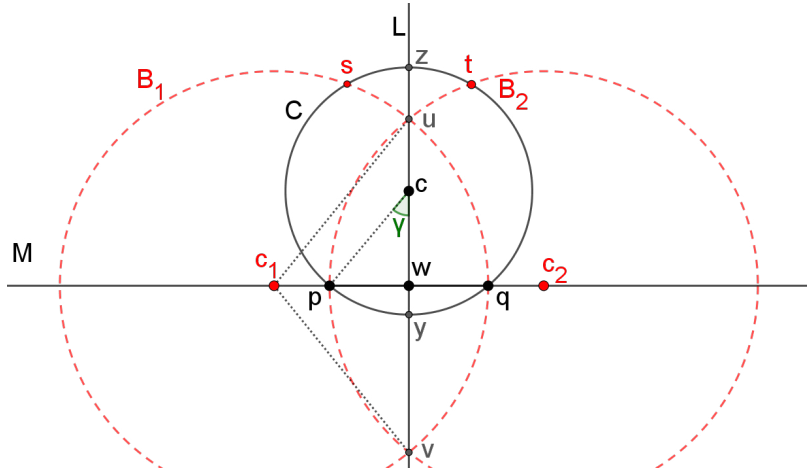


Fig. 7. Example of Lemma 1: $\text{lune}_\beta(p, q)$ with $\beta = 2.7$ where $\overline{pq}$ is a chord of a circle C with central angle $\alpha = 2\gamma = 80^\circ$.

A) Let w be a midpoint between p and q , L the perpendicular bisector of $\overline{pq}$ and z the intersection between L and $\widehat{pq}^+$ of $C = C(c, r)$. See Figure 7. With γ we denote the angle $\angle pcw = \frac{\alpha}{2}$. Then,

$$d(w, z) = \frac{1}{2}d(p, q) \left(\frac{1}{\sin(\gamma)} + \frac{1}{\tan(\gamma)} \right) = \frac{1}{2}d(p, q) \left(\frac{1 + \cos(\gamma)}{\sin(\gamma)} \right). \quad (16)$$

Let u be the intersection of the boundary of $\text{lune}_\beta(p, q)$ with L , such that $u \in H(M, c)$. Let $B_1 = B(c_1, r_\beta)$ and $B_2 = B(c_2, r_\beta)$ be defined at (1), (2) and (3). Then,

$$d(u, w)^2 = d(c_1, u)^2 - d(c_1, w)^2, \quad (17)$$

$$\Rightarrow d(u, w) = \frac{1}{2}d(p, q) \sqrt{2\beta - 1}. \quad (18)$$

Therefore, $d(w, z) \geq d(u, w)$ if and only if inequality (14) holds.

Assume (14) and let $s \neq p$ and $t \neq q$ be the intersections between the boundaries of B_1 and B_2 with $\widehat{pq}^+$, respectively. Then, $s \in \widehat{pz}^- \cup \{z\}$ and $t \in \widehat{qz}^- \cup \{z\}$, where $\widehat{pz}^-$ and $\widehat{qz}^-$ are in C . Thus,

$$B_1 \cap \widehat{pq}^+ \subseteq \widehat{ps}^- \quad \text{and} \quad B_2 \cap \widehat{pq}^+ \subseteq \widehat{qt}^-, \quad (19)$$

$$\Rightarrow (B_1 \cap B_2) \cap \widehat{pq}^+ \subseteq \widehat{ps}^- \cap \widehat{qt}^-, \quad (20)$$

$$\Rightarrow \text{lune}_\beta(p, q) \cap \widehat{pq}^+ \subseteq \emptyset. \quad (21)$$

Then, by convexity, $\text{lune}_\beta(p, q) \cap H(M, c) \subset B(c, r)$.

B) Now, suppose that y is the intersection between $\widehat{pq}^-$ and L , as Figure 7. Then,

$$d(w, y) = \frac{1}{2}d(p, q) \left(\frac{1 - \cos(\gamma)}{\sin(\gamma)} \right). \quad (22)$$

Let v be the intersection of the boundary of $\text{lune}_\beta(p, q)$ with L , such that $v \notin H(M, c)$. By symmetry about M , $d(v, w) = d(u, w)$, see equation (16). Therefore, $d(w, y) \leq d(w, v)$ if and only if

$$\left(\frac{1 - \cos(\gamma)}{\sin(\gamma)} \right)^2 + 1 \leq 2\beta. \quad (23)$$

The function $g(x) := \left(\frac{1 - \cos(x)}{\sin(x)} \right)^2 + 1$ is continuous, positive and strictly increasing in $x \in [0, \pi)$. Also, $g(\frac{\pi}{2}) = 2$. It means that inequality (23) holds for any $\beta \geq 1$. Then, $\widehat{pq}^- \subset \text{lune}_\beta(p, q)$. $\square$

The following is a technical lemma that helps in the proof of Lemma 3.

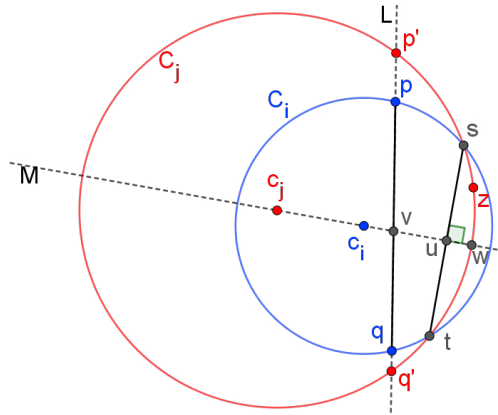


Fig. 8. Example of Lemma 2

Lemma 2. Let $\overline{pq}$ be a chord (not diameter) of $C_i = C(c_i, r_i)$ and S its circular segment. Let p' and q' be points in the line L that passes through p and q , such that $p', q' \notin B(c_i, r_i)$ and $\overline{pp'} \cap \overline{qq'} = \emptyset$.

If $C_j = C(c_j, r_j)$ passes through the points p', q' and z , where $z \in S$, then, c_i is in the interior of $\triangle pqc_j$.

Since z is inside C_i and p, q is not, then the boundaries of circles C_i and C_j must intersect at two points s and t . Note that if $p = p'$, then $p = s$, the same is true for $q = q'$ and t . The centers c_j and c_i lie on the bisector of $\overline{st}$, denoted with M . Let w be the point of intersection of the boundary of C_j with $M \cap S$ and u the midpoint of $\overline{st}$. Then c_j is on the ray of M that starts at c_i and does not contain w , that is, c_i is on the segment $\overline{c_j w}$. Let v be the intersection of $\overline{pq}$ with M . Since $\overline{pq}$ is not a diameter and $w \in S$, v is in the segment $\overline{c_i w}$, and $v \neq c_i, w$. Therefore, c_i is an interior point of the triangle $\triangle pqc_j$. For reference see Figure 8. $\square$

Lemma 3 is used in Theorem 1 to show that there is no vertex of T inside any $B(c_i, r_i)$.

Lemma 3. Let $T = \{T_1, T_2, \dots\}$ be a cyclic tiling, $C_i = C(c_i, r_i)$ the circle that circumscribes the tile T_i and V the vertex set of T .

Then, for $i = 1, 2, \dots$, $V \cap B(c_i, r_i) = \emptyset$.

Proof. By way of contradiction, suppose that $z \in V$ is in the interior of some circle $C_i = C(c_i, r_i)$. Since T is edge-to-edge then z is not in T_i . So, z must lie inside a circular segment S_i of $B_i = B(c_i, r_i)$ defined by an edge $\overline{pq}$ of T_i ; see Figure 9. Let T_j be the other tile that has $\overline{pq}$ as an edge. If z is a vertex of T_j inscribed in $C_j = C(c_j, r_j)$,

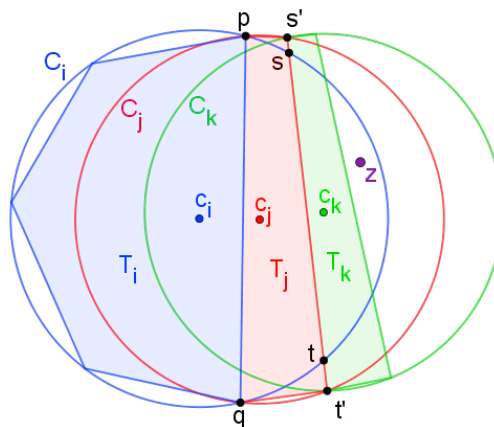


Fig. 9. Example of Lemma 3

by Lemma 2, c_i is in the interior of triangle $\triangle c_j p q \subset T_j$, which contradicts that T is a cyclic tiling since in this case the intersection of the interiors of two tiles is not empty. So, z is not a vertex of T_j and is in the interior of $S_i \setminus T_j$; see Figure 9.

Since z cannot be in T_j , if $S_i \subset T_j$, then we conclude that z does not exist. So suppose that $S_i \setminus T_j \neq \emptyset$, then it is formed by a finite set of circular segments of B_i . Let $S_j \subset S_i \setminus T_j$ be the circular segment that contains z and $\overline{st}$ the chord of C_i that determines it. The segment $\overline{st}$ determines an edge $\overline{s't'}$ of T_j . If T_k is the adjacent tile to T_j through $\overline{s't'}$ and z is a vertex of T_k , using Lemma 2 again in C_i and C_k we arrive at the contradiction that c_i is in interior of T_k . Therefore, z is not a vertex of T_k and must be in a circular segment $S_k \subset S_j \setminus T_k$, where S_k is a circular segment of C_i . Since C_i is a bounded region and the tiling is locally finite, repeating the above procedure we conclude that no such z can exist. $\square$

4. Conclusion

It has been found empirically that β -skeletons approximate well the street networks [5, 6, 7]. One question that arises from such empirical work is why it works so well. The interest in studying tilings starts to address such concern. So, proving that some tessellations are recovered by β -skeletons implies from a theoretical point of view that as far as street networks maintain certain regularity they will be nicely approximated by β -skeletons. Of course, street networks are not tilings and so we think that a better theoretical model is tessellations obtained from cyclic tilings by perturbing its vertex set. This is a more difficult task and it is work in progress.

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